

The IB core Extended Essay

Subject :

Physics Higher Level

Topic : An investigation on the factors affecting the storage capacity of optical discs

Research Question : How does the track spacing and the laser wavelength used by DVD and Blu-Ray formats affect their storage capacity ?

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Abstract :

An optical disc is a storage medium that reads data using optical technology. The objective of this report is to investigate the impact of optical disc technology on data storage, by focusing on a comparative study of DVD and Blue-Ray formats within the context of physics.

The research methodology involves an in-depth analysis of the physics principles underlying optical disc technology, including the properties of light, and diffraction. Through a comparative study, the research examines the differences in laser technology and data storage mechanisms employed in Blu-ray and DVD formats.

The laser technology and data storage mechanisms were investigated experimentally, with the use of diffraction grating experiment. The purpose of these investigations were to determine the laser wavelength and track spacing used by each optical disc.

In order to develop, the laser technology and data storage mechanisms, the data our diffraction experiments will be recorded and then worked out. This development will involve some physics bonded to some mathematical concept, such as, Trigonometry.

According to many sources, investigating these areas enable us to draw conclusions on the storage capacity of the optical disc. Which would therefore give us the opportunity to make ranking in terms of it. Overall, this EE research question combines knowledge of physics principles, technological advancements, and experimental techniques to explore the factors impacting the data storage of optical disc.

Word count = 228 words

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1. Introduction :

Over the course of the 2000s, our consumption of optical discs has undergone a number of changes. Available in different formats, including Blu-Ray and DVD, they are used for digital data backup and storage. They have played a big part in my childhood, when the consumption of optical discs underwent major changes, I was only a child. At the time, although I wondered about the subject, I didn't understand the success of Blue-Ray apart from the fact that it offered greater storage capacity. Today, having a more mature view of the world around me, I've been able to carry out my research in this field, providing me with clearer, more in-depth answers. In the course of my research, I learned that optical disc technology involves the use of a number of physics-related concepts, which explains why a Blu-Ray has a greater storage capacity. That's why I believed it was a good idea to do my physics EE on this subject, with the aim of understanding what scientific factors made Blue-Ray more qualitative than DVD.

Research Question : How does the track spacing and the laser wavelength used by DVD and Blu-Ray formats affect their storage capacity ?

To investigate the factors affecting their storage capacity, my essay will be divided into three parts:

- **Part A**, the laser wavelength
- **Part B**, the track spacing
- **Part C**, the length of the spiral track

In order to understand the effects it would be useful to know the following details informations about discs stated below.

2. Hypothesis :

As I said earlier, the advantage that Blu-Ray has over DVD is its storage capacity. To find the source of this phenomenon, we need to look at their differences, that are : their laser wavelength and track spacing. The fact that they differ in these areas leads me to believe that these factors are responsible for the storage capacity of optical discs.

3. Theory :

3.1. Mechanism inside DVD and Blu-Ray player :

In DVD and Blu-Ray player, a laser beam is responsible for reading data. The laser beam is directed towards the surface of the disc, where it encounters the pits and lands. These pits and lands act as obstacles, they are microscopic areas with different reflectivity making variations on the disc's surface, causing the laser beam to diffract.

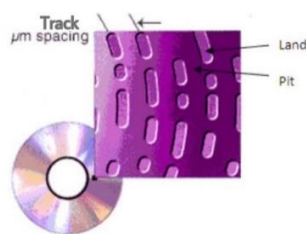


Figure 1 : Optical Disc's track spacing¹

¹ Fig. 1. Track pitches comparison of CD-R, Double Density CD -R and DVD. (n.d.). ResearchGate.
https://www.researchgate.net/figure/Track-pitches-comparison-of-CD-R-Double-Density-CD-R-and-DVD-3_fig1_261263539

As pits and lands acts as obstacles, then if a laser is directed on the disc's surface it will create an interference pattern. Thus leading to a physics' phenomenon called "Diffraction grating". Which is the spreading of light waves when they encounter a material composed of closely spaced parallel slits (grating element). In DVD and Blue-Ray player, the optical disc acts as the grating element when it interacts with its laser. Thus, producing a series of bright spots known as a interference pattern.

As directing a laser towards an optical disc or grating element creates an interference pattern, which from it an indicator of the storage capacity (track spacing) can be quantify. The track spacing can be worked using Trigonometry, as seen below.

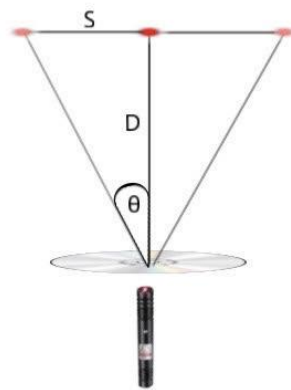


Figure 2 : Diffraction grating with laser²

It is known that $\tan\theta = \frac{\text{Opposite}}{\text{Adjacent}}$, which can be expressed in our case as :

$$\tan\theta = \frac{s}{D} \quad (1)$$

Where s is the fringe spacing, the distance between bright spots. And D is the screen-grating distance.

² vt.physics. (2019, October 19). Diffraction Grating Experiment: measuring wavelength | A Level Physics [Video]. YouTube. <https://www.youtube.com/watch?v=desLn3tMLcc>

Rearrange it, I can have the angle of diffraction :

$$\theta_D = \arctan \frac{s}{D} \quad (2)$$

Since the wave passes through a grating element, an equation³ can be used to express this diffraction phenomenon:

$$d \sin \theta = n \lambda \quad (3)$$

n = order of diffraction

λ = laser wavelength

d = grating spacing

θ = angle of diffraction

Then, we substitute equation (2) in (3) :

$$d \sin \left(\arctan \frac{s}{D} \right) = n \lambda \quad (4)$$

Putting "d" in subject we have :

$$d = \frac{n \lambda}{\sin \left(\arctan \frac{s}{D} \right)} \quad (5)$$

This interference patterns is supposed to take place inside the optical disc's player, where a photodetector detect these variations in reflection due to the pits and lands to then translate it into binary data. The photodetector will convert the intensity of the light into an electrical signal, enabling the retrieval of encoded data.

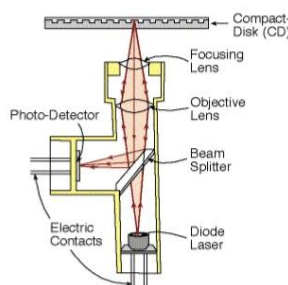


Figure 3 : The inference process inside a Blu-Ray/DVD player⁴

³ Diffraction on a CD — collection of experiments. (n.d.). <http://physicsexperiments.eu/1704/diffraction-on-a-cd>

⁴ Measure the Distance Between Tracks of CD and DVD[PDF]. uotechnology. (n.d.).

3.2. The spiral track on disc's surface :

Furthermore, the arrangement of these pits and lands make a spiral track to store data. It is a continuous, tightly wound path that begins in the disc's center and extends towards the outer edge.

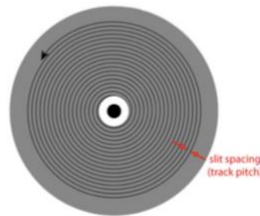


Figure 4 : The spiral track on the optical disc's surface⁵

A determination of the Spiral track's length is considered as a helpful tool, for insight on the storage capacity, because they use a tightly wound spiral track, optical discs maximize the amount of data that can be stored on their surfaces. The Spiral of Archimedes equation⁶ can be used to calculate the actual length of the spiral tracks on DVDs and Blu-rays :

$$L = \frac{\pi(R_{out}^2 - R_{in}^2)}{d} \quad (6)$$

L : length of spiral

R_{out} : outer radius

R_{in} : inner radius

d : track spacing

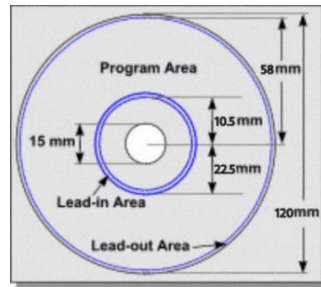
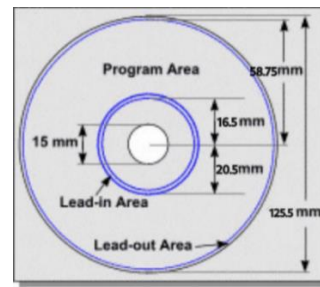
<https://uotechnology.edu.iq/dep-laserandoptoelec-eng/laboratory/4/laser%20application/exp3.pdf>

⁵Physical Optics[PDF]. Washington University in St. Louis. (n.d.).

https://web.physics.wustl.edu/introphys/Archives/SP15/PhysicalOptics_SP15.pdf

⁶ Pencheva, T., Gyoch, B., & Mashkov, P. (2010). Optical measurements upon compact discs in education in optoelectronics. Research Gate.

https://www.researchgate.net/publication/261263539_Optical_measurements_upon_compact_discs_in_education_in_optoelectronics

Figure 5 : Blu-Ray dimension⁷⁸Figure 6 : DVD dimension⁹

3.3. Optical Disc Laser Theory :

Another important point, is the fact that optical discs use laser with different wavelength. As DVD and Blue-Ray player use different laser of different colour, it therefore led to different wavelength. A blue laser is used in Blu-ray players, while a red one is used in DVD players. This difference in laser wavelengths has implications for data storage on optical discs. That's because smaller pits can be formed on the disc surface and more accurate focusing is made possible by shorter wavelengths., resulting in higher storage density. This factors collectively contribute to the overall data storage capacity and performance of the optical storage medium. In other words, a shorter wavelength allow the reading of lengthy spiral track on the optical disc's surface. This factors collectively contribute to the overall data storage capacity and performance of the optical storage medium¹⁰. This increased storage density enabling one optical disc to hold more data compared to another. Thus, the laser wavelength directly impacts the storage capacity and

⁷ Blu-ray replication – fermata.cz. (n.d.-b). <https://www.fermata.cz/our-offer/blu-ray-replication/?lang=en>

⁸ Where is TOC located on a CD. (n.d.) <http://www.herongyang.com/CD-DVD/CD-TOC-Where-Is-Located.html>

⁹ DVDs in standard size coloured DVD cases - Band CDs. (2024, January 25). Band CDs. <https://www.bandcds.co.uk/dvds/dvds-standard-size-coloured-dvd-cases/>

¹⁰ Into The Ordinary. (2017, September 7). How does blu-ray work? - LaserDisc, CD, DVD, Blu-ray explained [Video]. YouTube. <https://www.youtube.com/watch?v=H-jxTzFrnpq>

performance capabilities of optical storage systems like DVDs and Blu-ray discs. The wavelength of the laser used in DVDs and Blu-Rays would then be a necessary tool for determining the storage capacity of an optical disc. This is why the wavelength of laser diodes in optical disc player is found using a diffraction grating experiment, using equation (4).:

$$d \sin\left(\arctan \frac{s}{D}\right) = n\lambda \quad (4)$$

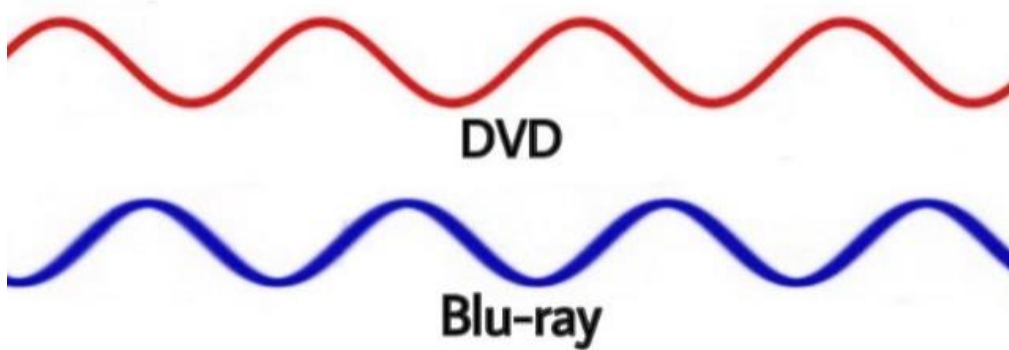


Table 7 : Laser wavelength used for each optical disc

4. Key Variables :

4.1. Independent Variables :

Independent Variables	How I'm going to measure them ?	Why am I measuring them ?
Laser wavelength	To determine the involvement of the laser wavelength in the DVD/Blue-Ray, I will use a diffraction experiment with a simulator. I'll vary the screen-grating distance, D, to collect as much data as possible. Using the following formula : $d\sin\theta = n\lambda$	Measuring the laser wavelength, will help us to have a clear insight of which optical discs can best read a long spiral track. Moreover, it will enable us to find the track spacing for each optical disc.
Track spacing	To calculate the track pitch, I will use DVD's and Blue-Ray's surface as a grating element to measure their track spacing. With the help of the same formula cited previously : $d\sin\theta = n\lambda$	The track spacing for each optical discs is inversely related to the length of the spiral track. Measuring the track spacing would therefore help us to affirm which of them has the best storage capacity.

4.2. Dependent Variables :

Dependent Variables	How I'm going to measure them ?	Why am I measuring them ?
The length of the Spiral Track	The Spiral track length will be found by using the Spiral of Archimedes equation. This equation will refer to the dimension of the optical disc and its track spacing (found from the wavelength).	The length of the Spiral track will be measured because it is a direct indicator of the storage capacity. A longer spiral track means there is more encoded data on the optical disc's surface. Meaning a higher storage capacity. This variable would therefore be useful for our comparison in terms of storage capacity.

4.3. Constant Variables :

Controlled Variables	Why am I controlling them ?	How am I going to control them ?
Temperature	By controlling temperature change, the intensity of the light source will have less variations, affecting the accuracy of the diffraction pattern.	I will conduct the experiment in a controlled environment, ensuring that the experiment is shielded from external temperature fluctuations.
Angle of Incidence	Different angles of incidence result in variations in the diffraction pattern. By controlling it, the same diffraction pattern will be consistently observed, allowing for reliable comparisons.	Use adjustable mounts for both, the grating element and light source. To securely hold the light source and the diffraction grating, thus allowing a reduction in variation of the angle of incidence.
Power of the laser beam	By controlling the power of the laser beam, unpredictable variations in the observed diffraction pattern can be avoided.	The intensity can be controlled directly by adjusting the power output of the laser. Many lasers have built-in power control mechanisms that allow you to regulate the intensity of the emitted light.

5. Methodology :

5.1. Experimental Setup :

- Blu-Ray disc
- DVD disc
- Red laser 5mW
- Blue laser 5mW
- Ruler ± 0.05 cm
- Grating element (500 lines per mm)
- Surface to project the diffracted pattern onto
- Mount for the diffraction grating and optical discs
- Dark room

5.2. Procedure :

To give some context, the procedures presented below were not applied in real life but through a simulator. These procedures are only there to give a clear idea of the steps to be taken to carry out this investigation.

5.2.1. Procedure for Part A : Laser wavelength

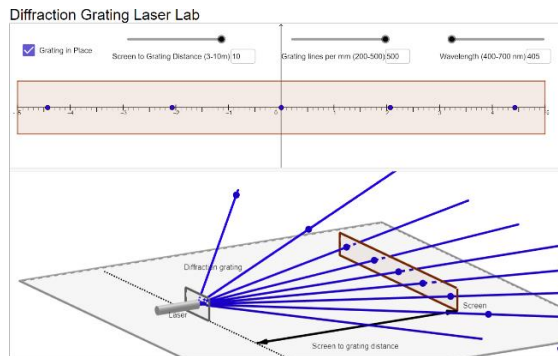


Figure 7 : Diffraction grating with the laser diode of Blu-Ray

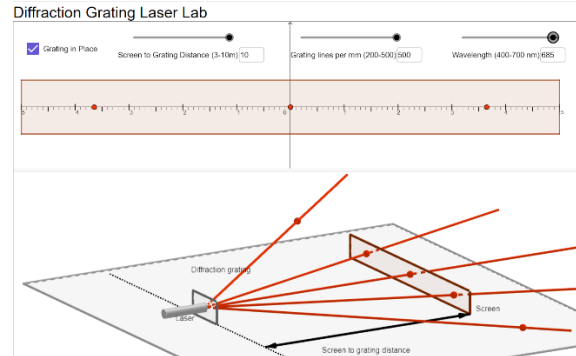


Figure 8 : Diffraction grating with the laser diode of DVD

1. Remove laser diode from the optical disc player.
2. Put the diffraction grating at a suitable distance from the screen to capture the diffracted pattern.
3. Place the diffraction grating and laser diode in a mount so that it is stable.
4. Position the laser diode of the optical disc so that the beam of light passes through the grating of 500 lines per mm.
5. An interference pattern will then be observed, displaying a series of bright spots of different intensities. From this, the laser wavelength can be determined from measurements of fringe spacing and screen-grating distance.
6. Afterwards, we can calculate the laser wavelength at different screen-grating distance to then make an average of the results. Thus, giving an accurate measure of the wavelength.

6.2.2. Procedure for Part B : The track spacing

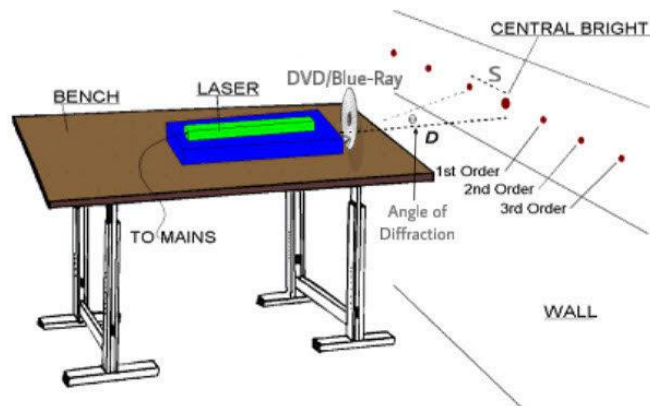


Figure 9 : Diffraction grating experiment with optical disc¹¹

1. Discard the top polycarbonate layer of the DVD and Blue-Ray.¹²
2. Place the optical disc on a flat surface with the shiny side facing up.
3. And fix it at a particular distance from the a white board (screen) such that the diffraction pattern would be displayed.
4. Position the light source, so that it is incident on the optical disc's surface.
5. Then, to get a clear view of the interference pattern, the experiment must be carried out in the dark.
6. From the interference pattern displayed, the fringe spacing (s) and the distance (D) can then be taken in order to calculate the track spacing.
7. Finally, the experiment is repeated at different distances to obtain an average of our results, for a more accurate measure of our final result.

¹¹ Hickson, I. (n.d.). Ian Hickson's laser Experiment. © Copyright 1998 by Ian Hickson.

<http://academia.hixie.ch/bath/laser/home.html>

¹² CD and DVD as diffraction gratings[PDF]. nnci. (n.d.). [https://nnci.net/sites/default/files/2020-02/CD DVD Diffraction%20grating Part%201 SG answers 0.pdf](https://nnci.net/sites/default/files/2020-02/CD%20DVD%20Diffraction%20grating%20Part%201%20SG%20answers%200.pdf)

6. Results and findings :

6.1. Raw Data Part A : Laser wavelength

Fringe spacing s/ m $\Delta s = \pm 0.01 \text{ m}$			Distance between the screen the grating element D/ m $\Delta D = \pm 0.1 \text{ m}$
1st Order	2nd Order	3rd Order	
1.70	1.6	/	5.0
1.90	2.8	/	5.5
2.05	/	/	6.0
2.22	/	/	6.5
2.40	/	/	7.0
2.60	/	/	7.5
2.75	/	/	8.0
2.90	/	/	8.5
3.10	/	/	9.0

Table 1 : Shows the variation in interference pattern with the laser diode of DVD when there is change in screen-grating distance

Fringe spacing s/ m			Distance between the screen the grating element D/ m $\Delta D = \pm 0.1$ m
$\Delta s = \pm 0.01$ m			
1st Order	2nd Order	3rd Order	
1.04	1.2	1.6	5.0
1.13	1.39	1.79	5.5
1.23	1.46	1.92	6.0
1.33	1.57	2.09	6.5
1.45	1.65	/	7.0
1.55	1.75	/	7.5
1.65	1.9	/	8.0
1.75	2	/	8.5
1.87	2.13	/	9.0

Table 2 : Shows the variation in interference pattern with the laser diode of Blu-Ray¹³ when there is change in screen-grating distance

¹³ OPhysics. (n.d.). <https://ophysics.com/l5b.html>

6.2. Raw Data Part B : Track spacing

Fringe spacing s/ m $\Delta s = \pm 0.001 \text{ m}$	Distance between the screen the grating element D/ m $\Delta D = \pm 0.001 \text{ m}$
0.425	0.200
0.524	0.250
0.629	0.300
0.734	0.350
0.839	0.400
0.944	0.450
1.049	0.500
1.154	0.550
1.259	0.600

Table 3 : Shows the variation in the interference pattern when there is change in screen-grating element, with DVD as grating element¹⁴ :

¹⁴ CD and DVD as diffraction gratings[PDF]. nnci. (n.d.). [https://nnci.net/sites/default/files/2020-02/CD DVD Diffraction%20grating Part%201 SG answers 0.pdf](https://nnci.net/sites/default/files/2020-02/CD%20DVD%20Diffraction%20grating%20Part%201%20SG%20answers%200.pdf)

Fringe spacing $s/ \text{ m}$ $\Delta s = \pm 0.001 \text{ m}$	Distance between the screen the grating element $D/ \text{ m}$ $\Delta D = \pm 0.001 \text{ m}$
0.761	0.120
0.824	0.130
0.888	0.140
0.951	0.150
1.015	0.160
1.078	0.170
1.142	0.180
1.205	0.190
1.268	0.200

Table 4 : Shows the variation in the interference pattern when there is change in screen-grating element, with Blu-Ray as grating element¹⁵

¹⁵ Into The Ordinary. (2017, September 7). How does blu-ray work? - LaserDisc, CD, DVD, Blu-ray explained [Video]. YouTube. <https://www.youtube.com/watch?v=H-jxTzFrnpg>

6.1. Processed data Part A : Laser wavelength

This section calculate the laser's wavelength for each optical disc's player using the data from Table 1 & 2. For instance, using the data from the first range of Table 1, the value of the DVD laser's wavelength will be found by following these steps :

1. Calculation of the angle of diffraction θ_D :

$$\begin{aligned}\theta_D &= \arctan \frac{s}{D} \\ &= \arctan \frac{1.7}{5}\end{aligned}\quad (2)$$

2. The slit separation on the grating element can be calculated using the equation (7) :

$$d = \frac{1}{N} \quad (7)$$

As the grating element has 500 lines per mm on it surface, as a result ;

$$\begin{aligned}d &= \frac{1}{500} \\ &= 2 \times 10^{-3} \text{mm}\end{aligned}$$

3. Calculation of the laser's wavelength :

From the equation (4) ;

$$d \sin\left(\arctan \frac{s}{D}\right) = n\lambda \quad (4)$$

Substitute the values ;

$$\begin{aligned}(2 \times 10^{-6}) \sin\left(\arctan \frac{1.7}{5}\right) &= (1)\lambda \\ \lambda &\approx 644 \times 10^{-9} \text{nm}\end{aligned}$$

I therefore repeated these steps for each variation in distance (screen-grating), giving several values for the wavelength. After collecting the values, an average of the values will be taken for more accuracy.

Laser wavelength $\lambda/\times 10^{-9}\text{m}$									
DVD	644	653	647	646	648	655	650	645	651
Blu-Ray	407	402	402	401	406	404	404	403	407

Table 5 : Shows the variation of the laser wavelength of Blu-Ray/DVD when there is change in screen-grating distance (D) in Part A

The average of the laser's wavelength in DVD and Blue-Ray, will be calculated using this formula :

$$\text{Average wavelength} = \frac{\sum \lambda}{n} \quad (8)$$

In DVD ;

$$\begin{aligned} \text{Average wavelength} &= \frac{644+653+647+646+648+655+650+645+651}{9} \\ &= 648.78 \text{ nm} \\ &\approx 649 \text{ nm} \end{aligned}$$

In Blue-Ray ;

$$\begin{aligned} \text{Average wavelength} &= \frac{407+402+402+401+406+404+404+403+407}{9} \\ &= 404 \text{ nm} \end{aligned}$$

6.2. Processed Data Part B : Track spacing

This section aims to calculate the track spacing (d) on the optical discs' surface, These would be done from the data on the Table 3 & 4, by following the step below:

1. Apply the equation (5):

$$d = \frac{n\lambda_{avg}}{\sin(\arctan \frac{s}{D})} \quad (5)$$

$$d = \frac{(1)649 \times 10^{-9}}{\sin(\arctan \frac{0.43}{0.20})}$$

$$\approx 0.716 \mu\text{m}$$

6.2.1. Table for the variation of track spacing of Blu-Ray/DVD when there is change in distance (D)

track spacing d/ $\times 10^{-6}$ m									
DVD	0.716	0.720	0.719	0.720	0.719	0.720	0.719	0.719	0.719
Blu-Ray	0.409	0.409	0.409	0.409	0.409	0.409	0.409	0.408	0.409

Table 6 : Shows the variation of the track spacing of Blu-Ray/DVD at each change in screen-grating distance (D) in Part B

The same process as earlier, is then applied on the track spacing to calculate the average of the track spacing in DVD and Blue-Ray, using this formula :

$$\text{Average track spacing} = \frac{\sum d}{n} \quad (9)$$

DVD ;

$$d_{avg} = \frac{0.716+0.720+0.719+0.720+0.719+0.720+0.719+0.719+0.719}{9}$$

$$= 0.719 \mu\text{m} \approx 0.72 \mu\text{m}$$

Blu-Ray ;

As there is surprisingly no variation the track spacing will still be $\approx 0.41 \mu\text{m}$,

$$d_{avg} = 0.41 \mu\text{m} .$$

6.3. Processed Data Part C : The length of spiral track

From the values of track spacing, the Archimedes equation can then be used for DVD and Blu-Ray from their dimension in Figure 6.

For DVD ;

- Radius of lead-in area $R_{in} = 16.5 \text{ mm}$, $R_{out} = 20.5 \text{ mm}$
- Radius of program area $R_{in} = 20.5 \text{ mm}$, $R_{out} = 58.75 \text{ mm}$
- Radius of lead-out area $R_{in} = 58.75 \text{ mm}$, $R_{out} = 59.25 \text{ mm}$

Which implies that :

- For lead in area (5) :

$$L = \frac{\pi(20.5^2 - 16.5^2) \times 10^{-6}}{0.719 \times 10^{-6}}$$

$$= 646.67 \text{ m}$$

- For program area (5) :

$$L = \frac{\pi(58.75^2 - 20.5^2) \times 10^{-6}}{0.719 \times 10^{-6}}$$

$$= 13244.99 \text{ m}$$

- For lead out area (5) :

$$L = \frac{\pi(59.25^2 - 58.75^2) \times 10^{-6}}{0.719 \times 10^{-6}}$$

$$= 257.79 \text{ m}$$

The summation of these lengths will give us the length of the spiral track :

$$\text{The length of spiral track} = 646.67 + 13244.99 + 257.79$$

$$= 14149.45 \text{ m}$$

For Blu-Ray ;

- Radius of lead-in area $R_{in} = 10.5 \text{ mm}$, $R_{out} = 22.5 \text{ mm}$
- Radius of program area $R_{in} = 22.5 \text{ mm}$, $R_{out} = 58 \text{ mm}$
- Radius of lead-out area $R_{in} = 58 \text{ mm}$, $R_{out} = 58.5 \text{ mm}$

Then ;

- For lead in area (5) :

$$L = \frac{\pi(22.5^2 - 10.5^2) \times 10^{-6}}{0.41 \times 10^{-6}}$$

$$= 3034.319 \text{ m}$$

- For program area (5) :

$$L = \frac{\pi(58^2 - 22.5^2) \times 10^{-6}}{0.41 \times 10^{-6}}$$

$$= 21897.28 \text{ m}$$

- For lead out area (5) :

$$L = \frac{\pi(58.5^2 - 58^2) \times 10^{-6}}{0.41 \times 10^{-6}}$$

$$= 446.336 \text{ m}$$

Therefore ;

$$\begin{aligned}
 \text{The length of spiral track} &= 3034.319 + 21897.28 + 446.336 \\
 &= 3034.319 + 21897.28 + 446.336 \\
 &= 25377.94 \text{ m}
 \end{aligned}$$

6.4. Verification of our data:

As with every experiment, uncertainties and errors do occur, so it's important to quantify them to see how accurate our data is.

6.4.1. Uncertainty propagation

Part A : The calculation of the laser's wavelength was done by using the fringe spacing (s) and the distance (D). But since they themselves are subject to uncertainties, they propagated some in the laser wavelength ;

$$\Delta s = \pm 0.01 \quad \text{and} \quad \Delta D = \pm 0.1$$

The equation (4) states ;

$$d \sin\left(\arctan \frac{s}{D}\right) = n\lambda \quad (4)$$

Rearrange it ;

$$\lambda = \frac{d \sin\left(\arctan\left(\frac{s}{D}\right)\right)}{n} \quad (10)$$

Which implies, from law of uncertainty, that ;

$$\frac{\Delta\lambda}{\lambda} = \frac{\Delta \sin \left(\arctan \left(\frac{s}{D} \right) \right)}{\sin \left(\arctan \left(\frac{s}{D} \right) \right)} \quad (11)$$

The absolute uncertainty of the wavelength is therefore ;

$$\Delta\lambda = \lambda \left(\frac{\Delta \sin \left(\arctan \left(\frac{s}{D} \right) \right)}{\sin \left(\arctan \left(\frac{s}{D} \right) \right)} \right) \quad (12)$$

Where ;

$$\begin{aligned} \Delta \sin \left(\arctan \left(\frac{s}{D} \right) \right) &= \frac{1}{2} (\text{Max} - \text{Min}) \\ &= \frac{1}{2} \left(\sin \left(\arctan \left(\frac{s+0.01}{D-0.1} \right) \right) - \sin \left(\arctan \left(\frac{s-0.01}{D+0.1} \right) \right) \right) \end{aligned} \quad (13)$$

As a result ;

$$\Delta\lambda = \lambda \left(\frac{\frac{1}{2} \left(\sin \left(\arctan \left(\frac{s+0.01}{D-0.1} \right) \right) - \sin \left(\arctan \left(\frac{s-0.01}{D+0.1} \right) \right) \right)}{\sin \left(\arctan \left(\frac{s}{D} \right) \right)} \right) \quad (14)$$

This formula can be applied on the values from Table 1 & 2 to find the absolute uncertainty for each wavelength. For example :

For the first range of Table 2 ;

$$\begin{aligned} \Delta\lambda &= 404 \times 10^{-9} \left(\frac{\frac{1}{2} \left(\sin \left(\arctan \left(\frac{1.04+0.01}{5-0.1} \right) \right) - \sin \left(\arctan \left(\frac{1.04-0.01}{5+0.1} \right) \right) \right)}{\sin \left(\arctan \left(\frac{1.04}{5} \right) \right)} \right) \\ &\approx 1.302 \times 10^{-8} \text{m} \\ &\approx 13.02 \text{ nm} \end{aligned}$$

This can then be repeated for all the wavelength calculated, as seen below ;

Approximate absolute uncertainty of wavelength $\Delta\lambda$ / nm									
DVD	21.38	20.06	18.82	17.83	17.01	16.35	15.64	15.02	14.59
Blu-Ray	13.02	11.98	11.10	10.35	9.708	9.155	8.671	8.243	7.864

Table 7 : Shows the variation of absolute uncertainty of the wavelength with the screen-grating distance (D) for Blu-Ray/DVD in Part A

Part B : This part focuses instead on calculating the track spacing. The calculations show that the fringe spacing (s), the distance (D) and the wavelength (λ) are important parameters for calculating the track spacing. But they are also subject to uncertainties propagating on the track spacing ;

$$\Delta s = \pm 0.001 \quad \Delta D = \pm 0.001 \quad \Delta \lambda_{\text{DVD}} = \pm 17.43 \quad \Delta \lambda = \pm 10.01$$

From equation (5) ;

$$d = \frac{n\lambda_{\text{avg}}}{\sin(\arctan \frac{s}{D})} \quad (5)$$

Using law of uncertainty ;

$$\frac{\Delta d}{d} = \frac{\Delta \lambda_{\text{avg}}}{\lambda_{\text{avg}}} + \frac{\Delta \sin(\arctan \frac{s}{D})}{\sin(\arctan \frac{s}{D})} \quad (15)$$

Rearrange it ;

$$\Delta d = d \left(\frac{\Delta \lambda_{\text{avg}}}{\lambda_{\text{avg}}} + \frac{\Delta \sin(\arctan \frac{s}{D})}{\sin(\arctan \frac{s}{D})} \right) \quad (16)$$

Using equation (9) and put it in uncertainty form ;

$$\frac{\Delta\lambda_{\text{avg}}}{\lambda_{\text{avg}}} = \frac{\Delta\Sigma\lambda_n}{\Sigma\lambda_n} \quad (17)$$

Reorganize it ;

$$\Delta\lambda_{\text{avg}} = \lambda_{\text{avg}} \left(\frac{\Delta\Sigma\lambda}{\Sigma\lambda} \right) \quad (18)$$

Now, we substitute our values;

DVD ;

$$\begin{aligned} \Delta\lambda_{\text{avg}} &= 649 \left(\frac{21.38+20.06+18.82+17.83+17.01+16.35+15.64+15.02+14.59}{644+653+647+646+655+650+645+651} \right) \\ &= 17.36 \text{ nm} \end{aligned}$$

Blue-Ray ;

$$\begin{aligned} \Delta\lambda_{\text{avg}} &= 404 \left(\frac{13.02+11.98+11.10+10.35+9.70+9.15+8.67+8.24+7.86}{407+402+402+401+406+404+404+403+407} \right) \\ &= 10.01 \text{ nm} \end{aligned}$$

It is then possible to work out the value of the absolute uncertainty of track spacing at each range in the table, using the equation (). This can be illustrated for the 1st Range of DVD in Table 3, as seen below :

$$\begin{aligned} \Delta d &= d \left(\frac{\Delta\lambda_{\text{avg}}}{\lambda_{\text{avg}}} + \frac{\frac{1}{2}(\sin(\arctan(\frac{s+0.001}{D-0.01})) - \sin(\arctan(\frac{s-0.001}{D+0.01})))}{\sin(\arctan(\frac{s}{D}))} \right) \quad (19) \\ \Delta d &= 0.716 \times 10^{-6} \left(\frac{17.36 \times 10^{-9}}{649 \times 10^{-9}} + \frac{\frac{1}{2}(\sin(\arctan(\frac{0.425+0.001}{0.20-0.01})) - \sin(\arctan(\frac{0.425-0.001}{0.20+0.01})))}{\sin(\arctan(\frac{0.425}{0.20}))} \right) \\ \Delta d &= 0.026 \text{ } \mu\text{m} \end{aligned}$$

This process is then repeated several times in a row for each range.

Approximate absolute uncertainty of track spacing $\Delta d/ \mu\text{m}$									
DVD	0.03	0.02	0.02	0.02	0.02	0.02	0.02	0.02	0.02
Blu-Ray	0.01	0.01	0.01	0.01	0.01	0.01	0.01	0.01	0.01

Table 8 : Shows the variation of absolute uncertainty in track spacing at each change in screen-grating distance (D) for Blu-Ray/DVD in Part B

The absolute uncertainty of d_{avg} can then be found for each disc from this equation :

$$\Delta d_{\text{avg}} = d_{\text{avg}} \left(\frac{\Delta \Sigma d_n}{\Sigma d_n} \right) \quad (20)$$

DVD ;

$$\begin{aligned} \Delta d_{\text{avg}} &= 0.72 \left(\frac{0.03+0.02+0.02+0.02+0.02+0.02+0.02+0.02+0.02}{6.48} \right) \\ &= 0.02 \mu\text{m} \end{aligned}$$

Blue-Ray ;

$$\begin{aligned} \Delta d_{\text{avg}} &= 0.41 \left(\frac{0.01+0.01+0.01+0.01+0.01+0.01+0.01+0.01+0.01}{3.69} \right) \\ &= 0.01 \mu\text{m} \end{aligned}$$

From these calculations, we can account for the effect of the uncertainties of each quantity on the result by using the percentage uncertainty :

$$\text{Percentage Uncertainty} = \frac{\text{Uncertainty}}{\text{Measured Value}} \times 100 \quad (21)$$

DVD laser wavelength ;

$$\begin{aligned} \text{Percentage Uncertainty} &= \frac{17.43}{649} \times 100 \\ &= 2.685\% \end{aligned}$$

Blue-Ray laser wavelength ;

$$\begin{aligned} \text{Percentage Uncertainty} &= \frac{10.01}{404} \times 100 \\ &= 2.477\% \end{aligned}$$

DVD track spacing ;

$$\begin{aligned} \text{Percentage Uncertainty} &= \frac{0.02}{0.72} \times 100 \\ &= 2.77\% \end{aligned}$$

Blue-Ray track spacing ;

$$\begin{aligned} \text{Percentage Uncertainty} &= \frac{0.01}{0.41} \times 100 \\ &= 2.44\% \end{aligned}$$

Through these calculations, it can be seen that the percentage uncertainty is relatively small, making our measurement reliable. Furthermore, the percentage uncertainties are similar, which therefore suggests that our measurements were obtained using similar methods. Thus fostering again the idea that our measurements are reliable.

6.4.2. Error propagation :

The measure of error propagation can be done using this equation:

$$\text{Percentage Error} = \left| \frac{\text{Measured value} - \text{True Value}}{\text{True Value}} \right| \times 100 \quad (22)$$

This equation will be used for the laser's wavelength and the track spacing of each optical disc, by using the average values calculated previously and their standard value :

- DVD : 650 nm and 0.74 μm ¹⁶
- Blu-Ray : 405 nm and 0.35 μm

DVD ;

Wavelength (λ) ;

Track spacing (d) ;

$$\% \text{ Error} = \left| \frac{649 - 650}{650} \right| \times 100$$

$$\% \text{ Error} = \left| \frac{0.72 - 0.74}{0.74} \right| \times 100$$

$$= 0.1538 \%$$

$$= 2.7 \%$$

Blu-Ray ;

Wavelength (λ) ;

Track spacing (d) ;

$$\% \text{ Error} = \left| \frac{404 - 405}{405} \right| \times 100$$

$$\% \text{ Error} = \left| \frac{0.41 - 0.35}{0.35} \right| \times 100$$

$$= 0.2469 \%$$

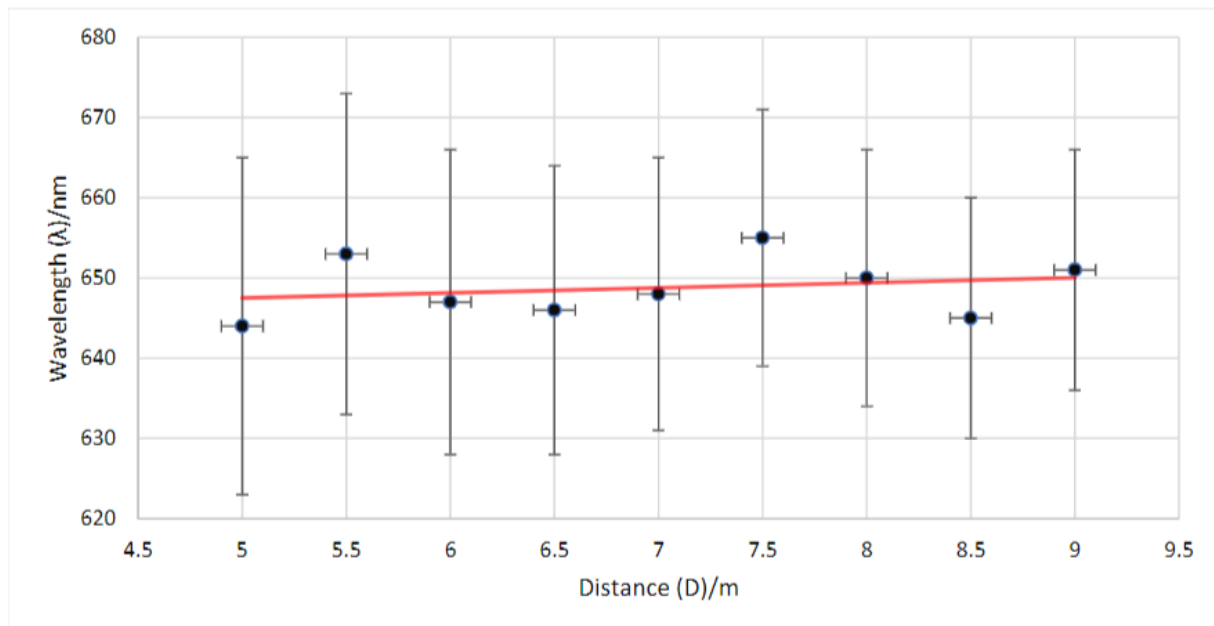
$$= 17.1429 \%$$

On the contrary of percentage uncertainty, the percentage error nous montre que nos valeurs ne sont pas toutes accurate, surtout pour le track spacing of Blu-Ray. It has a percentage error of 17.14 %, which is somewhat significant. This means there is lack of accuracy in this area caused by unaccounted factors.

¹⁶ CD and DVD as diffraction gratings[PDF]. nnci. (n.d.). [https://nnci.net/sites/default/files/2020-02/CD DVD Diffraction%20grating Part%201 SG answers 0.pdf](https://nnci.net/sites/default/files/2020-02/CD%20DVD%20Diffraction%20grating%20Part%201%20SG%20answers%200.pdf)

6.5. Graph obtained from Data :

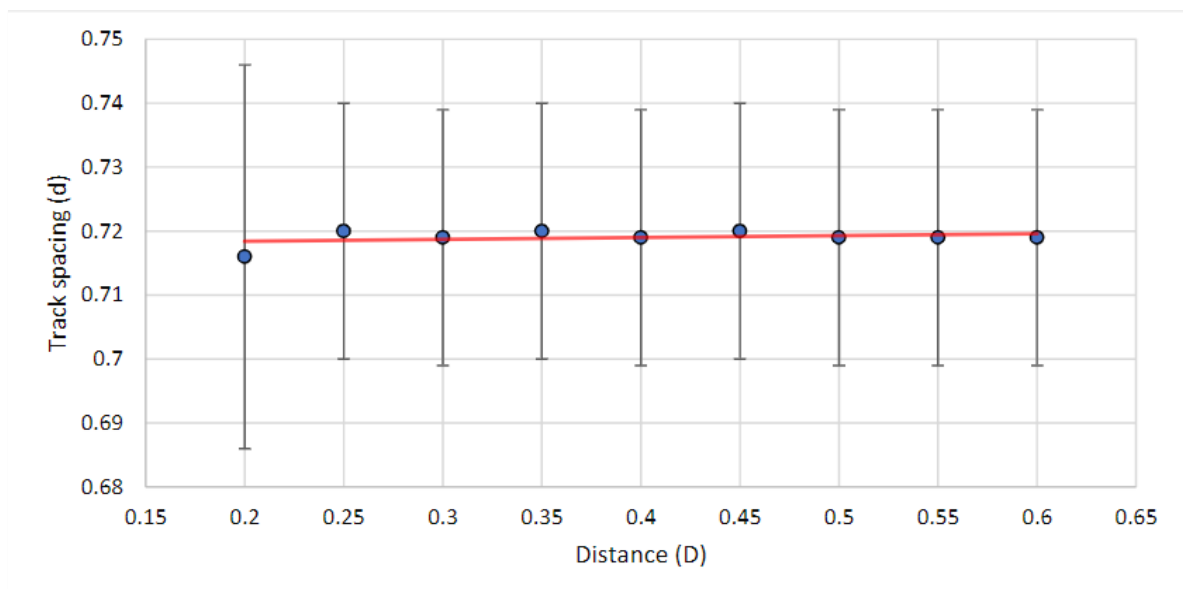
6.5.1. Graph 1 for wavelength versus distance D for DVD :



All of these graphs introduced are correlated with our previous calculations. The first one, clearly show that the DVD laser wavelength always acts at a range between 655 and 644 nm, which is around 650 nm. The vertical axis represents the wavelength of the lasers and the horizontal axis represents the distance between the screen and the grating element. In order to represents the relationship between wavelength and the screen-grating distance, the graph used a linearized with error bars. This error bars indicate the variability associated with data points. They represent the variability or dispersion of the data around a central value, such as the mean value. Through the diagrams it can be seen that as the distance D increases there is a reduction of error bars in a graph. This phenomenom is caused by the increase in sample size (increase in distance D). With a larger sample size, the variability of the data tends to decrease. As more data points are included in the analysis, the estimates become more precise, leading to narrower error bars. Increasing the sample size can provide more reliable and accurate measurements, reducing the uncertainty associated with the data. From this

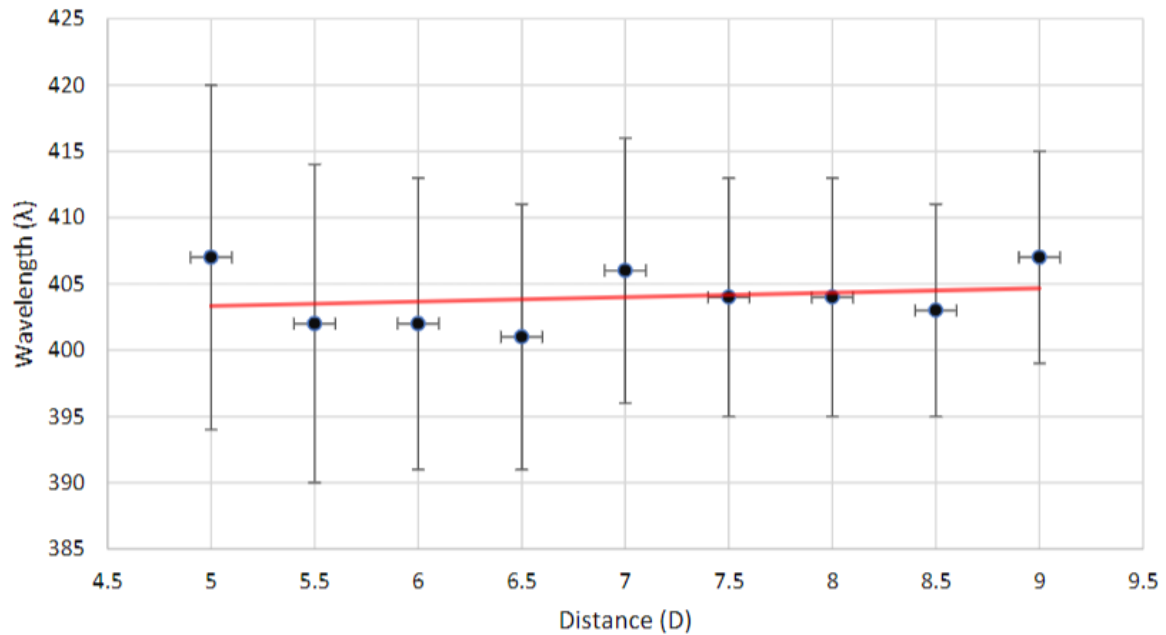
background, it can be understood the data points with large sample size on our graph, are the most accurate one. Referring to this, we can say that this graph supports the idea that a 650 nm laser is commonly used in DVD players. As the linearized curve reach a value of 650 nm, when the distance is at its highest.

6.5.2. Graph 2 for track spacing versus distance D for Blu-Ray from Part B



The second graph, displays the variation of the track spacing with screen-grating distance on DVD, stating that it acts in range between 0.718 and 0.720. The vertical axis represents the track spacing on the optical disc surface and the horizontal axis represents the distance between the screen and the grating element. The relationship between track spacing and the screen-grating distance, is again represented by the use of an linearized curve. This linearized curve shows that the track spacing on DVD surface is likely to be 0.72 μm.

6.5.3. Graph 3 for wavelength versus distance for Blue-Ray



From the third graph introduced above, it can be seen that the Blue-Ray laser has various values for its wavelength, following a range between 407 and 401. Through the diagram, it can clearly be seen that the laser's wavelength in DVD is longer than the Blue-Ray. This graph illustrates by following exactly the same structure as the first graph, that the wavelength of the Blu-ray laser operates at a wavelength around 405 nm, which is constantly shorter than the wavelength of the DVD laser, no matter the change in distance.

Furthermore, a graph of Blu-Ray track spacing is not plotted as there is no variation because it is constantly equal to $0.41 \mu\text{m}$.

7. **Conclusion :**

7.1. How the optical disc technology may impact the storage capacity ?

From our data developed, conclusions can be drawn explaining why the storage capacity of DVDs and Blu-rays differs.

First of all, it is already known that the laser technologies that DVDs and Blu-ray discs use to read and write data are responsible of the difference in storage capacity, because it is said that a shorter laser wavelength allows greater data density on the disc. Blu-ray discs use a shorter blue-violet laser with a wavelength of about equal to 405 m, whereas DVDs use a red laser with a wavelength of about 650 nm (nanometers). Since a Blu-Ray player has a shorter wavelength, it therefore enables the storage of more data.

Furthermore, the idea that Blu-Ray has a higher storage capacity is still supported by track spacing measurement. As it is said that the tighter track spacing of an optical disc allows for more data to be packed into a given area of the disc's surface. From our results, the DVDs have a track spacing of around 0.74 micrometers (μm) while on Blu-ray is around 0.41 micrometers. The track spacing on Blu-ray is significantly narrower than that of DVD, which therefore prove that Blu-Ray has a higher storage capacity.

Finally, these two parameters explored, lead to difference in the length of the spiral track. As explored in our previous calculations, we have been able to see that the length of spiral track of Blu-ray is longer than the one of DVD (Blu-Ray : 25377.94 m > DVD : 14149.45 m). All these factors therefore support the idea that Blu-ray has higher storage capacity than DVD.

These points can be easily supported from the data on the storage capacity of each optical disc. DVDs can typically hold up to 4.7 GB of data for single-layer discs. This capacity is smaller compared to the storage capacities of Blu-ray discs, which is approximately 25 GB.¹⁷

8. Evaluation

Despite the fact that this investigation is reliable, I did face some limitations in implementing my analysis. Initially, I wanted to do the experiments in person. But due to the cost of materials, I resigned myself to that. I then carried out my analyses using a simulator and reliable data to develop my analysis points. Moreover, the main obstacle I encountered was calculating the track spacing of each optical disc. During these calculations I had to make estimates based on data I had obtained from research, for the track spacing of DVD and especially for Blu-Ray. This explains why it had a high level of percentage error. Added to the fact that my estimates caused errors, the data on which I based myself comes from an experiment. Which means that our data was already subjected to error. During diffraction grating experiment with optical discs, there are several potential causes of errors :

- 1) **Alignment Issues** : Despite the use of a mount, some vibrations may arrived, thus impacting the proper alignment of the diffraction grating.
To fix this issue the grating should have been aligned more correctly with the incident light source, thus enabling error reduction.

¹⁷ *Introduction to DVD technology*[PDF]. wordpress. (n.d.).
<https://etcitcse.files.wordpress.com/2012/01/chap-1-cd-dvd1.pdf>

- 2) **Human Error** : Errors can also arise from mistakes made during the experiment, incorrect readings, or inconsistent technique. Carelessness, or lack of experience can contribute to human errors. These errors may have been reduced if proper procedures had been followed.
- 3) **Imperfection in the Grating** : The diffraction grating itself may have imperfections that can impact the accuracy of the measurements. Some irregularities in the spacing between the lines can introduce errors in the observed diffraction pattern. I think the use of high-quality gratings would help to minimize these sources of error.

On the contrary, the wavelength calculations were not affected by errors as they were based on a simulator for measurements. Simulator reduce potential errors as it uses:

- 1) **Controlled Environment** : Simulators provide exact control over a range of conditions because they run in a controlled environment. This carefully controlled environment removes extraneous elements that can cause measurement errors. Simulators can produce results that are more consistent by removing variables from the real world.
- 2) **Known Inputs** : Known inputs are the foundation upon which simulators operate. These models are created with data and well-established scientific principles in mind. The simulator can produce accurate outputs if it receives accurate data. The use of known inputs therefore helps ensure that the simulation accurately represents the phenomenon being studied.

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10. Appendix

Part A : Laser wavelength

θ / Angle of Diffraction/in degrees									
Range	1 st	2 nd	3 rd	4 th	5 th	6 th	7 th	8 th	9 th
DVD	18.73	19.02	18.85	18.85	19.07	18.91	18.96	18.79	18.96
Blu-Ray	11.75	11.50	11.58	11.56	11.70	11.67	11.65	11.63	11.73

Part B : Optical disc track spacing

θ / Angle of Diffraction/in degrees									
Range	1 st	2 nd	3 rd	4 th	5 th	6 th	7 th	8 th	9 th
DVD	64.80	64.49	64.50	64.50	64.51	64.51	64.52	64.52	64.52
Blu-Ray	81.03	80.16	80.19	80.22	80.13	80.16	80.18	80.20	80.21